IPCC Tier 2 Livestock GHG Uncertainty Calculator

User Guide

CIAT / CGIAR Alliance — ClimateActionNetZero

Contents

Alliance Bioversity & CIAT

Global Methane Hub

Version 1.0

Developed under the Global Methane Hub initiative

1 Overview

1.1 What this tool does

National GHG inventories based on IPCC Tier 2 methods require not only a central estimate of emissions but also a quantification of their uncertainty. This tool automates that process for livestock (cattle) using Monte Carlo simulation:

1. You supply your country-specific Tier 2 activity data and emission-factor inputs, each with an uncertainty range.
2. The tool randomly samples from the uncertainty distributions you have defined (thousands of times) and propagates those samples through the full IPCC Tier 2 equation chain (Eq. 10.1–10.34).
3. It reports the resulting 95% confidence interval around your emission estimate, identifies which parameters drive the most uncertainty (sensitivity analysis), and formats the results for submission in the format of IPCC 2006 Vol. 1 Ch. 3 Table 3.3.

The approach follows IPCC Approach 2 (Monte Carlo) as described in IPCC 2006 Vol. 1 Ch. 3.

1.2 Who this tool is for

This tool is designed for national inventory practitioners who:

- Are already conducting or have recently completed a Tier 2 cattle GHG inventory (enteric fermentation, manure management, direct and indirect N_2O from pasture deposition)
- Have at least rough estimates of the uncertainty in their key activity data and emission factors
- Need to report the uncertainty range for inclusion in national inventory reports

The tool assumes basic familiarity with IPCC Tier 2 methodology. Users do not need prior experience with Monte Carlo methods, but a conceptual understanding of probability distributions is helpful.

1.3 Emission sources covered

IPCC Category	Source
3.A.1	Enteric fermentation — cattle (CH_4)
3.A.2	Manure management — cattle (CH_4)
3.A.2	Manure management — cattle (N_2O direct)
3.A.2	Manure management — cattle (N_2O indirect)
3.C.4	Direct N_2O from pasture/range/paddock deposition
3.C.5	Indirect N_2O from pasture/range/paddock deposition

2 Preparing your data

2.1 What you need before you start

Before opening the tool, assemble the following:

1. **Activity data**: For each animal sub-category, the population (N), body weight (BW), weight gain (WG), milk yield (for dairy), milk fat %, digestible energy (DE%), manure management system (MMS) fractions, and other Tier 2 inputs.

2. **Uncertainty ranges**: For each parameter, an estimate of its uncertainty — typically either:

   - A percentage uncertainty (e.g. ±20% representing the half-width of a 95% confidence interval), or
   - Explicit lower and upper bounds.

3. **Manure management fractions** (if running manure N_2O): The percentage of manure allocated to each management system (pasture, solid storage, liquid, etc.). If you are uncertain about the allocation itself, the Manure_Management tab also has optional `lower_fraction` / `upper_fraction` / `distribution_fraction` columns — fill them and the tool will sample each MMS within its range, renormalising every Monte Carlo iteration so the simplex (per-group rows sum to 100%) is preserved. The sampled values surface as `fraction_<mms>` in the sensitivity tornado so MMS allocation can show up as an influential driver. Leave blank to keep `fraction_pct` deterministic (the default, matching the IPCC Inventory Software).

If you do not yet have uncertainty estimates, see Section 2.4 for guidance on obtaining them.

You only need the parameters used by the emission sources you tick. The tool's pre-run check is source-aware: if you select only the methane sources (enteric, manure CH_4), it will not ask you to fill the N_2O parameters, and vice versa. So a methane-only run never blocks on blank manure-N_2O cells. The managed-storage N_2O quantities (EF3 per system, Frac_GasMS, Frac_LeachMS) live in the Manure_Management tab — fill them there, not in the Parameters tab.

2.2 AI-assisted data preparation (optional, free)

If your raw inventory data lives in your own Excel or CSV files with column names that do not match the template, you can use the free **GMH Uncertainty Translator** on `claude.ai` to do the column-mapping and unit-conversion work for you. The Translator is a small AI assistant — loaded with the parameter catalogue, the template schema, and ten worked mapping examples — that you set up once inside your own free Claude.ai account and then reuse for every inventory. It is free, requires no installation, and your data stays inside your own account.

One-time setup (about 2 minutes). The Resources tab offers a downloadable `translator_kit.zip` containing a system-instructions file plus four small knowledge files. After signing up for a free Claude.ai account, you create a Project in claude.ai (Projects → Create project), paste the contents of `system_instructions.md` into the Project's *Instructions* field, then drag the four knowledge files (`param_catalogue.md`, `template_schema.md`, `mapping_examples.md`, `questionnaire.md`) into the Project's *Files* panel. That is the whole setup — you only do it once. Step-by-step screenshots are in the Getting Started guide available from the Resources tab.

Workflow (each inventory). The chat with the Translator has two distinct inputs you provide: a **pre-flight questionnaire** that you *paste as text* into the chat, and your **raw data file(s)** that you *attach* when Claude prompts you. The questionnaire is not a separate upload — it is a one-page form whose contents you copy-paste into the chat box, so Claude knows about your inventory up front and can skip its warm-up questions.

1. Download the **pre-flight questionnaire** (`.docx`) from the Resources tab and fill it out — country, year, IPCC version, your sub-categories, manure systems, data fields, uncertainty source (about 2 minutes).
2. Open your Translator Project on claude.ai and click *Start a new chat.*
3. In the chat box, **paste the filled questionnaire as your first message**. This is text you copy-paste, not a file you attach. Claude will summarise back what it understood and then ask you to upload your data.
4. **Attach your raw data file(s)** using the paperclip icon in the chat input box (`.xlsx`, `.csv`, or even a PDF / screenshot of a table).
5. Claude proposes a column mapping with confidence flags; confirm ambiguities (typically 2–5 short questions, e.g. *"Is your column* `weight` *body weight or mature weight?"*).
6. Download the produced `filled_template_for_app.xlsx` and upload it in Tab 1 (Data Input).

When to use. The Translator is most useful when (a) your data is in a non-standard shape, (b) your column names differ from the template (e.g. `cattle_pop`, `live_wt_lb`), (c) you have data in inconsistent units (lbs vs kg, fractions vs percentages), or (d) you want IPCC defaults filled in for the parameters you do not measure.

Quality control. The Translator does column-mapping, not analysis. Always (a) spot-check the key numbers (population, body weight, milk yield) by comparing with your original file, and (b) review the QA/QC tab after upload — any value the Translator filled from an IPCC default is flagged in amber there.

2.3 Complete parameter reference

The table below lists all parameters accepted by the tool. Required parameters depend on the emission sources you are running. Parameters marked *n/a* in the IPCC default column must be supplied from your national data. The tool will auto-fill missing optional parameters with IPCC defaults and flag them in Tab 1 (see Section 3.2).

Table 1: Complete parameter reference — all Tier 2 inputs accepted by the tool.

Code	**Definition**	**Unit**	**IPCC default**	**Suggested uncertainty**	**Suggested distribution**	**IPCC reference**
Activity data						
N	Number of animals in this sub-category	head	*n/a*	±10%	Normal	Eq. 10.1
BW	Average live body weight (also TAM = Typical Animal Mass)	kg	275	±15%	Normal	Eq. 10.3 / 10.6
MW	Mature (adult) body weight — used in NEg growth equation	kg	300	±10%	Normal	Eq. 10.6
WG	Average daily weight gain — set 0 for non-growing adults	kg/day	0	±30%	PERT	Eq. 10.6
Milk	Daily milk yield per *lactating* animal (not sub-category average; tool multiplies by pct_pregnant internally). Set 0 for non-dairy sub-categories.	kg/head/day	4.0	±20%	Normal	Eq. 10.8
Fat	Fat content of milk	%	4.0	±10%	Normal	Eq. 10.8
pct_pregnant	Fraction of females in this sub-category pregnant during the year, 0–1 — includes pregnant heifers that have not yet calved. Aliases `pct_calving` and `pct_lactating` are accepted on upload.	fraction	0.60	±20%	Beta	2019 Refinement Vol.4 Ch.10 §10.5; Eq. 10.13

Continued on next page...

...continued from previous page

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
MilkPR	Protein content of milk — feeds the milk-N term in Eq. 10.33 (N retention rates for cattle, where the 6.38 milk-protein-to-N conversion is defined). The 2019 Refinement gives the relation %MilkPR = 1.9 + 0.4×%Fat (so 4% fat → 3.5% protein); illustrative values are tabulated in Annex 10A.1 / 10A.2 / 10A.3.	%	3.3	±10%	Normal	Eq. 10.33; Annex 10A.1–10A.3
Energy coefficients						
DE	Digestible energy as % of gross energy — typical range 45–75%	%	55.0	±15%	Normal	Eq. 10.14
Cfi	Maintenance energy coefficient — depends on sex and lactation status	MJ/day/kg$^{0.75}$	0.386	±30%	PERT	Table 10.4
Ca	Activity coefficient for locomotion energy	dimensionless	0.17	±30%	Triangular	Table 10.5
C	Growth coefficient for NEg equation — depends on sex (female 0.8, castrate 1.0, bull 1.2)	dimensionless	0.8	±30%	Triangular	Eq. 10.6
Cp	Pregnancy coefficient — 0.10 for pregnant animals	dimensionless	0.10	±10%	Beta	Table 10.7
hours	Daily working hours — set 0 when animals do no work (typical for dairy/beef cattle inventories)	hours/day	0	±20%	PERT	Eq. 10.11

Continued on next page...

…continued from previous page

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Tw	Mean daily temperature in winter (°C) — Cfi cold-climate adjustment. Leave blank or set 20 to disable.	°C	20	±25%	Normal	Eq. 10.2
Enteric fermentation						
Ym	Methane conversion factor (% of gross energy in feed converted to CH_4)	%	6.5	±8%	PERT	Table 10.12
Manure management — CH_4						
Bo	Maximum CH_4 producing capacity of manure	m^3 CH_4/kg VS	2019R: 0.13 (Other regions, low productivity cattle); 0.24 (intensive dairy NA/W.Europe); 2006 Africa = 0.10	±20%	PERT	Table 10.16(a) (Updated)
ASH	Ash content of manure (fraction of dry matter)	fraction	0.08	±25%	PERT	Eq. 10.24
UE	Urinary energy as fraction of gross energy	fraction	0.04	±25%	PERT	Eq. 10.24

Continued on next page…

…continued from previous page

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
CP	Crude protein content of diet — used to estimate nitrogen excretion	%	10.0	±15%	Normal	Eq. 10.32
Manure management — N_2O (indirect emission factors)						
EF4	N_2O EF for atmospheric N deposition (used in Eq. 10.27)	kg N_2O-N/(kg NH_3-N + NO_x-N)	2006: 0.010; 2019R aggregated: 0.010; wet 0.014; dry 0.005	asymmetric	PERT	Eq. 10.27; Ch.11 Table 11.3
EF5	N_2O EF for N leaching/runoff (used in Eq. 10.29)	kg N_2O-N/kg N	2006: 0.0075; 2019R: 0.011	asymmetric	PERT	Eq. 10.29; Ch.11 Table 11.3
*The managed-storage N_2O quantities EF3(S), Frac_GasMS and Frac_LeachMS are entered **per manure-management system** in the Manure_Management tab (not in the Parameters tab), because each value is specific to a storage system. The tool reads them from there.*						
Manure management — N_2O (pasture/range/paddock)						
EF3_PRP	N_2O emission factor for dung/urine deposited on pasture (used in Ch.11 Eq. 11.1)	kg N_2O-N/kg N	2006: 0.02; 2019R $EF3_{PRP,CPP}$ aggregated: 0.004; wet 0.006; dry 0.002	asymmetric	PERT	Ch.11 Eq. 11.1; Table 11.1

Continued on next page…

…continued from previous page

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Frac_GasPRP	Fraction of N volatilised from dung/urine on pasture (used in Ch.11 Eq. 11.9)	fraction	0.21 (cattle)	asymmetric	PERT	Ch.11 Eq. 11.9; 2019 Table 11.3
Frac_LeachPRP	Fraction of N lost through leaching from pasture deposition (used in Ch.11 Eq. 11.10)	fraction	2006: 0.30; 2019R: 0.24 (wet) / 0 (dry)	asymmetric	PERT	Ch.11 Eq. 11.10; 2019 Table 11.3

Note on asymmetric uncertainty: Parameters flagged "asymmetric" use explicit lower/upper bounds from the IPCC tables rather than a symmetric ±% formula. The template pre-fills these bounds automatically from the IPCC 2019 Refinement tables. *Note on* `pct_pregnant`: this is the canonical name (matches IPCC Vol.4 Ch.10 Eq. 10.13's "% females pregnant" — includes pregnant heifers that have not yet calved). Legacy column names `pct_lactating` and `pct_calving` are automatically renamed to `pct_pregnant` on upload, so existing templates continue to work without edits.

2.4 Understanding the input structure

The tool uses an Excel template with a specific structure. Download the template from the **Data Input** tab (Tab 1) by clicking "Download Excel template".

2.5 The Inventory_Metadata sheet

The first sheet captures who and where the inventory is for. Two cells deserve a note:

- **Country**: free text (e.g. `Zimbabwe`). Used in the report header — not used for any lookup.
- **Continental region**: a **dropdown** (`africa / asia / europe / americas / oceania / global`). This drives the QA tab's BW deviation check, which compares your body-weight value against the IPCC Vol.4 Ch.10 Annex Tables 10A.1 / 10A.2 / 10A.3 default for your continent. Pick the continent your inventory's animal sub-categories are in. If you genuinely do not know or are running a hypothetical example, select `global`.

Legacy templates: if you uploaded a template downloaded before this region field was split out (i.e. only the old single `Country / region'' cell), the parser maps common country names (Zimbabwe, India, Brazil, ...) to the right continent automatically. Anything unrecognised falls back to \texttt{global} (no false BW warnings; the QA message will just sayglobal'`'). Re-downloading the current template is recommended but not required.

The Parameters sheet has the following key columns:

Column	Description	Example
`cattle_type`	Broad cattle category	`dairy`, `non_dairy`
`aggregation_level`	Production system or region name	`Highland smallholder dairy`
`sub_category`	Animal sub-group	`cows`, `heifers`, `calves_female`
`parameter`	Parameter code (see Table 1)	`N`, `BW`, `Ym`, `DE`, `Bo`
`mean`	Central estimate (point value from your inventory)	`450` (BW in kg)
`distribution`	Uncertainty distribution shape	`normal`, `pert`, `lognormal`
`uncertainty_pct`	Symmetric ±% half-width of 95% CI	`20`
`lower`	Lower bound (alternative to uncertainty_pct)	`360`
`upper`	Upper bound	`540`
`param_type`	IPCC framing	`activity_data` or `coefficient`
`unit`	Unit of measurement	`kg`, `%`, `fraction`

Important: Each row is one parameter for one sub-category. A country with two production systems and five sub-categories each would have approximately 12–15 parameters × 10 sub-categories = 100–150 rows.

2.6 Defining sub-categories

Sub-categories should match your national inventory. Typical sub-categories include:

- Dairy: `cows` (lactating), `dry_cows`, `heifers`, `calves`, `bulls`
- Non-dairy beef: `breeding_cows`, `growing_males`, `heifers`, `calves`, `oxen`

Each sub-category needs its own rows for all relevant parameters. The values of `cattle_type`, `aggregation_level`, and `sub_category` together identify each group uniquely — they must be spelled consistently across all rows for the same group.

2.7 Obtaining uncertainty estimates

For activity data (N, BW, milk yields):

- **Survey-based estimates**: If populations are estimated from livestock surveys, the survey sampling error provides a direct estimate of uncertainty. The 95% confidence interval from a survey translates directly to the `lower` and `upper` bounds (or to `uncertainty_pct` if symmetric).
- **Expert elicitation**: Ask the data provider to give a range within which they are 95% confident the true value falls.
- **IPCC defaults**: IPCC 2019 Refinement Chapter 11 provides uncertainty ranges for key emission factors. Use these as the starting point if no country-specific data is available.

For emission factors and coefficients (Ym, Bo, DE, MCF, etc.):

- **Literature review**: If you used a country-specific value from a published study, the study's reported confidence interval or standard deviation can be used.
- **IPCC default uncertainty ranges**: IPCC 2019 Refinement (Vol.4 Ch.10 / Ch.11) gives lower / upper bounds for several emission factors (e.g. $EF_{3,PRP}$, Ym, MCF). The template pre-fills these where IPCC publishes them; where IPCC does not publish a CI, the template falls back to ranges from Penman *et al.* (2000) and Monni *et al.* (2007) — these are clearly distinguished in the template footnotes.
- **Expert elicitation**: Ask the expert who provided the value to estimate the plausible range.

If you genuinely have no information about a parameter's uncertainty, a reasonable default for emission factors is ±20–40%. For population data with census-based estimates, ±5–15% is typical.

2.8 Choosing a distribution

Distribution	Shape	Best for
Normal	Symmetric bell curve	Activity data (N, BW) with symmetric measurement errors
PERT	Smooth bounded curve	Emission factors with a min, mode, and max
Triangular	Peaked triangle	Clear most-likely value with known hard limits
Uniform	Flat (equal probability)	Conservative fallback when only bounds are known
Lognormal	Right-skewed	Strictly positive values with right-skewed uncertainty (Bo, EFs)
Beta	Bounded $[0, 1]$	Fractions (pct_calving, Cp)

Recommended defaults:

- Use **Normal** for population (N) and body weight (BW) — typically measured with roughly symmetric errors
- Use **PERT or Lognormal** for emission factors (Ym, Bo, MCF, EF3–EF5, Frac parameters) — bounded below at zero with right-skewed uncertainty
- Use **Beta** for fractions bounded to [0, 1] (pct_calving, Cp)
- Use **Uniform** as a conservative fallback when you only know the plausible range

3 Loading your data (Tab 1 — Data Input)

3.1 Using the example datasets

Two built-in example datasets are included to let you explore the tool without uploading your own data:

- **Country X (hypothetical dairy)**: A fictional East African highland dairy system — 12 parameters, dairy/cows, including a synthetic 5-year time series.
- **Country Y (hypothetical pastoral)**: A fictional semi-arid pastoral beef system — 11 parameters, non-dairy/breeding cows, including a synthetic 5-year time series.

Select either from the "Country / Example Data" dropdown. The tool loads the data immediately and you can proceed directly to Tab 2.

3.2 Uploading your own data

Step 1 — Choose an IPCC Guidelines version. Before downloading the template, select the IPCC version your inventory follows:

- **IPCC 2006**: Uses the eight original manure management systems (MMS). Choose this if your national inventory is based on the 2006 Guidelines.
- **IPCC 2019 Refinement**: Adds four additional MMS types (anaerobic digester/biogas, aerobic treatment, solid storage covered/compacted, burned for fuel) and uses updated emission factors and N excretion equations from the 2019 Refinement. Choose this for inventories aligned with the 2019 Refinement.

The MMS dropdown in the downloaded template is automatically filtered to show only the manure systems valid for the version you selected.

Step 2 — Download and fill in the template.

1. Click "Download Excel template" to get the pre-formatted workbook
2. Fill in the **Parameters** sheet following the structure in Section 2.3
3. Fill in the **Manure_Management** sheet if you are running manure N_2O sources
4. *(Optional)* Fill in the **Parameter_TimeSeries** sheet if you want correlations computed automatically from historical data
5. Save the file as .xlsx and upload it using the file picker

Step 3 — Review upload notifications.

After a successful upload, a green notification confirms how many parameters were loaded. Any warnings appear as amber notifications.

Missing parameters and auto-fill: If your upload does not include all required parameters, the tool will:

- Still load and display all parameters that *were* supplied — you can review them immediately in Tab 2 and Tab 3.

- Automatically fill in missing parameters using IPCC default values (see the "IPCC default" column in Table 1).
- Show a **yellow warning notification** listing exactly which parameters were auto-filled and with what values (e.g. "*3 parameters auto-filled from IPCC defaults: Ym = 6.5, Bo = 0.10, DE = 55.0*").
- Flag those parameters as **Missing** (purple/grey) in the QA/QC tab (Tab 2).

You should review whether the IPCC defaults are appropriate for your country context and, where possible, supply country-specific values before your final simulation run.

Common upload issues:

- `cattle_type` must be `dairy` or `non_dairy` (lowercase, no spaces)
- Distribution names must match exactly: `normal`, `pert`, `triangular`, `uniform`, `lognormal`
- If `uncertainty_pct` is left blank, both `lower` and `upper` must be filled
- `param_type` must be `activity_data` or `coefficient`

4 Reviewing data quality (Tab 2 — QA/QC)

4.1 What the QA/QC checks do

The QA/QC tab runs automated checks on your parameter values and flags any that are potentially problematic. Five statuses are possible:

Status	Colour	Meaning
Pass	Green	Value is within ±50% of the IPCC default (Vol.4 Ch.10 / Ch.11)
Warn	Amber	Value deviates 50–200% from the IPCC default — check and document
Fail	Red	Value deviates >200% from the IPCC default — likely a data entry error
Missing	*Grey*	Parameter not found in upload; auto-filled with the IPCC default (Vol.4 Ch.10 / Ch.11)

Scope of the benchmark-deviation check. The deviation check is applied **only to BW (body weight)** — the one parameter for which the tool keeps a defensible continental IPCC table lookup (Annex Tables 10A.1 dairy / 10A.2 non-dairy / 10A.3 buffalo). Milk, MW, DE, Ym, Bo no longer fire benchmark warnings — the 2019 Refinement publishes these by production system rather than continent, so any single mid-point comparison was misleading. They are still pre-filled with IPCC defaults when missing (flagged *Missing*), they just no longer raise *Warn* purely for differing from a heuristic mid-point.

Cross-sheet sub-category-key check. If a Parameters-sheet `sub_category` (e.g. `DINT_heif`) has no exact match in the Manure_Management sheet but *is* close to exactly one Manure_Management sub-category (e.g. `DINT_heifer`, edit distance ≤ 2), the tool auto-matches them and emits a *Warn* row labelled `sub_category_auto_match` so you can verify the substitution. If multiple candidates match, the run is blocked with a clear error.

4.2 What to do for each status

Pass: No action needed. The value is within ±50% of the IPCC default for the parameter (Vol.4 Ch.10 / Ch.11).

Warn: The value deviates 50–200% from the IPCC default **(BW only)**, or another non-fatal QA condition fired (e.g. asymmetric-bounds reminder for $EF_{3,PRP}$/Frac_GASM_PRP/Frac_LEACH_PRP, sub-category auto-match). Possible actions:

- Confirm the value matches your country context
- If it is country-specific and well-justified, document the source in your inventory report
- If you intended to use a climate-disaggregated default, check IPCC Vol.4 Ch.11 Tables 11.1 / 11.3 for the wet/dry value
- If it is an error, correct it in your Excel file and re-upload

Fail: The value deviates more than 200% from the IPCC default. Most often this means a data-entry error (e.g. a live weight of 50,000 kg, a DE% of 150%). Correct the value before proceeding.

Missing: A required parameter was not found in the upload. The tool has auto-filled it with the IPCC default value from PARAM_CATALOGUE (the value and the IPCC equation/table source are shown in the Tab 1 notification). Review whether the default is appropriate for your country context and, where possible, replace it with country-specific data.

4.3 Documenting deviations

For any Warn or Info flags where you are deliberately using a country-specific value that deviates from the default, document the justification in your national inventory report. IPCC expert reviewers will look for this documentation.

5 Reviewing uncertainty distributions (Tab 3 — Uncertainty)

5.1 The parameter table

Tab 3 shows all parameters loaded from your data as an editable table. Each row shows:

- The parameter name, unit, and current mean value
- The distribution type currently assigned
- The uncertainty_pct or lower/upper bounds
- A visual preview of the distribution shape

You can edit values directly in the table. Clicking a cell opens an edit field.

5.2 Quick-set buttons

Two quick-set buttons are available for rapid exploration:

- **Set all activity data to Normal ±15%**: Sets N, BW, and other activity data parameters to a Normal distribution with ±15% uncertainty. Click again to undo.
- **Set all coefficients to PERT**: Sets emission factors and other coefficients to PERT distribution. Click again to undo.

These are useful for a first exploratory run. Refine individual parameters before your final analysis.

6 Correlations (Tab 4 — Correlations)

6.1 Why correlations matter

If multiple parameters tend to move together (e.g. larger animals also tend to produce more milk), ignoring this relationship underestimates the uncertainty in the total emission estimate. For most Tier 2 analyses, ignoring correlations (the default "No correlations" setting) is a reasonable simplifying assumption. IPCC Approach 2 guidance recommends including known correlations where they exist.

6.2 Quick guide — which option do I pick?

- Do you have ≥ 5 years of national time-series data in your upload? $\rightarrow$ **From template (auto, time-series)**.
- No time series, but you want to capture well-known biological linkages (BW $\leftrightarrow$ MW, Milk $\leftrightarrow$ Fat, DE $\leftrightarrow$ Ym, ...)? $\rightarrow$ **Structural defaults (expert-elicited)**. This is the recommended pick for single-year inventories without historical data.
- You have your own correlation matrix from a study or expert elicitation? $\rightarrow$ **Advanced — manual entry**.
- None of the above? $\rightarrow$ leave **No correlations** (the default; conservative and defensible).

Options you can't pick are greyed out. If a mode would have nothing to work with — e.g. From template'' when your \texttt{Parameter_TimeSeries} sheet is empty, orAdvanced — manual entry' ' before you've uploaded a CSV — the radio choice appears greyed out with a short note explaining why. The tool also blocks the Run button with an explicit error if a no-data mode is somehow selected.

6.3 What change in the result should I expect?

The central emission estimate is essentially unchanged regardless of which mode you pick — correlations preserve marginal means by construction. What changes is the **width of the 95% uncertainty interval**. As a rule of thumb, on a typical single-year cattle inventory:

- **No correlations** $\rightarrow$ the baseline. All parameters drawn independently.
- **Structural defaults** $\rightarrow$ widens the 95% uncertainty interval by roughly **3 to 10%**. Most of this comes from the DE $\leftrightarrow$ Ym pair (the only cross-block link in the preset, also the strongest at $\rho = -0.5$). The effect is intentionally modest because the preset only links 7 pairs out of $\sim$78 possible across the IPCC equation chain.
- **From template (auto, time-series)** $\rightarrow$ varies with the data. A rich, varied historical series (multiple parameters genuinely co-moving year-on-year) can produce effects similar to or larger than the structural defaults. A sparse or mostly static series will produce a much smaller effect because the matrix collapses to near-identity once constant columns are dropped.
- **Advanced — manual entry** $\rightarrow$ entirely depends on what you upload. Strong correlations

on parameters with wide uncertainty bounds can move the interval by 20% or more (in either direction).

- **Block-structured EF correlation** (right-hand card) at $\rho = 0.3$ on one block → a further few-percent widening on top of any activity-data correlation. At $\rho = 0.5$ (the slider maximum), roughly double that.

If you select correlations and the result barely changes, that is usually expected, not a bug — the IPCC equation chain multiplies around 13 parameters together, which dilutes the effect of any single correlated pair. See the Methodology document for the underlying mathematics.

6.4 Automatic correlations from time-series data

If you included a Parameter_TimeSeries sheet in your upload, the tool can compute historical correlations automatically:

1. On Tab 4, select **From template (auto, time-series)** under Activity Data Correlations.
2. The tool computes the **Spearman** rank correlation matrix of the historical parameter data after first-difference detrending (year-on-year changes, which strips out shared long-run growth) and samples using **Iman–Conover restricted pairing** — the method explicitly cited by IPCC V1 Ch3 §3.2.3.2 (p. 26). Iman–Conover is distribution-free and reproduces the target rank correlation exactly, regardless of the marginal shapes.
3. Inspect the heatmap that appears below the controls to confirm correlations are plausible.

6.5 Manual correlation matrix

If you prefer to specify correlations manually:

1. Select **Advanced — manual entry** under Activity Data Correlations.
2. Upload a square, symmetric CSV matrix of correlation coefficients (-1 to $+1$, diagonal = 1). The value you enter for cell (i, j) is the **Spearman rank correlation** between parameters i and j, which the tool preserves through Monte Carlo per IPCC Vol.1 Ch.3 §3.2.3.2.
3. The tool verifies positive-definiteness and projects onto the nearest positive-definite correlation matrix if needed; you are warned if any entry shifts by more than 0.01.

Downloadable starting templates. Two CSV templates appear inside the **Advanced — manual entry** panel and remove the need to construct the matrix from scratch:

- **Download blank matrix template** — a square matrix with all parameter names already in place as row and column headers, the diagonal set to 1, and every off-diagonal cell set to 0. Open it in Excel, fill the cells you want correlated, save as CSV, and upload via the **Upload correlation matrix (.csv)** button.
- **Download matrix with example values** — the same matrix but pre-filled with the seven structural-defaults pairs (BW ↔ MW, BW ↔ WG, Milk ↔ Fat, Milk ↔ BW, Milk ↔ DE,

DE ↔ CP, DE ↔ Ym). This matrix is **identical** to what the in-app *Structural defaults* preset applies internally — it is a useful starting point if you want to tweak just one or two pairs.

Parameter-name rules. The templates ship with the canonical parameter names already in the first column and the header row (`N`, `BW`, `MW`, `WG`, `Milk`, `Fat`, `pct_pregnant`, `DE`, `Cfi`, `Ca`, `C`, `Cp`, `hours`, `CP`, `Ym`, `Bo`, `ASH`, `UE`, `EF3_PRP`, `EF4`, `EF5`, `Frac_GASM_PRP`, `Frac_LEACH_PRP`, `MilkPR`, `Tw`). **Do not rename rows or columns** — names that don't match a catalogue parameter are silently ignored when the matrix is applied to the simulation (no error message; the correlation just doesn't take effect). Legacy aliases such as `pct_calving` are resolved on the *Parameters-sheet* upload only; on the correlation matrix they need to be the canonical names listed above.

Missing parameters are assumed to have zero correlation. If your uploaded matrix omits a parameter that the simulation uses (for example, you upload a small matrix covering only the parameters you want correlated), every pair involving the omitted parameter is treated as independent — the simulation runs and uses the correlations you did provide, but it does not invent values for the missing ones. Equivalent rule for header typos: a misspelled row/column name is treated the same as a missing parameter, so the correlation simply does not take effect. This is intentional graceful degradation, not an error, so the easiest way to verify your matrix is being used as you expect is to use the downloadable templates above (which ship with all canonical names already in place).

6.6 Structural defaults (expert-elicited)

A built-in preset of structural defaults is available. It applies the following well-documented pairs, drawn from the IPCC equations and the livestock literature. *These values are expert-elicited; the IPCC publishes no numerical correlation coefficients for these pairs.* Hover any non-zero cell in the heatmap for the per-pair source citation.

Parameter A	Parameter B	ρ	Source
BW	MW	+0.50	Growth-curve relation (Brody 1945); IPCC Eq 10.3. Set to 0.5 rather than 0.85 to reflect that BW comes from the national census while MW is usually a breed reference — different sources, so the cross-source correlation is closer to 0.5 than 0.85.
BW	WG	+0.40	IPCC Eq 10.6 joint estimation (NRC 2001 energy-balance studies, published range 0.30–0.50). Sign is system-dependent; +0.40 reflects well-fed adult dairy/beef systems.
Milk	Fat	−0.30	Milk dilution effect, dairy genetics literature (Wilmink 1987; VandeHaar 1998); within-herd literature −0.20 to −0.40.
Milk	BW	+0.30	High-producing dairy breeds are physically larger (Holstein ∼650 kg & ∼30 kg milk/day vs Jersey ∼450 kg & ∼20 kg). NRC Dairy 2001; Stallings & Knowlton 2013.
Milk	DE	+0.30	Higher feed digestibility ⇒ more nutrient availability ⇒ higher milk yield, especially in concentrate-fed dairy. NRC Dairy 2001 Ch.2.
DE	CP	+0.50	Forage-quality co-variation across feed types (NRC 2001; INRA; Feedipedia).
DE	Ym	−0.50	IPCC 2019 Refinement Eq 10.21 (Ym decreases with DE) + meta-analysis (Niu et al. 2018 GCB; Hristov et al. 2013). The only cross-block (activity-data ↔ coefficient) pair, and the dominant mover on the headline.

All other pairs are set to zero.

Derivation notes. A previously included N ↔ BW = 0.30 pair was removed (no defensible cross-country source); DE ↔ Ym is set to −0.50 rather than −0.40 (cross-diet meta-analyses); Cfi ↔ Ca is set to 0.30 rather than 0.60 (Cfi and Ca are independent inputs in IPCC Eq 10.3 / 10.4); BW ↔ MW is set to 0.50 rather than 0.85 (typical mixed-source data provenance); two cross-group dairy pairs (Milk ↔ BW and Milk ↔ DE) were added; Milk ↔ pct_pregnant is not included (sign ambiguous).

6.7 Coefficient (emission factor) correlations

The right-hand card on Tab~4 controls correlations *between* the per-head IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, …). Two modes are available:

- **No EF correlations** (default) — coefficients are sampled independently. This is the IPCC Approach 2 default and is appropriate when each emission factor comes from its own study.
- **Block-structured EF correlation** (recommended when you do want non-zero EF correlations) — the IPCC coefficients are grouped into three blocks, each with its own within-block ρ slider in $[0, 0.5]$:
  - *Energy-equation coefficients* (Cfi, Ca, C, Cp, Ym) — rumen-fermentation studies (IPCC Eq 10.3 / 10.16 / 10.21 family).
  - *Manure-CH_4 coefficients* (Bo, MCF, ASH) — BMP assays and lagoon-temperature studies.
  - *Manure-N coefficients* (EF3, EF4, EF5, Frac_GASMS, Frac_LEACH, UE) — NH_3 / N_2O volatilisation studies.

  Cross-block correlation is zero — these three literatures are independent. $\rho = 0.3$ is a moderate assumption capturing shared measurement bias within a block.

Note: a single-ρ "Uniform EF correlation" option was removed in the May 2026 audit. Applying one ρ to all coefficients implies that rumen-fermentation, BMP-lagoon and NH_3 volatilisation studies share a common bias, which is statistically indefensible.

7 Running the simulation (Tab 5 — Simulate)

7.1 Settings

Before clicking Run, configure:

Number of iterations: The number of Monte Carlo samples.

- **1,000**: Quick test run to check the model runs correctly. Do not use for reporting.
- **10,000**: Minimum for final reporting purposes. Tail percentiles (2.5th/97.5th) are reliable.
- **25,000–30,000**: Recommended for final reporting, especially when correlations are enabled or you have many sub-categories.

GWP version: Choose between AR5 (CH_4 GWP = 28, N_2O = 265) and AR6 (CH_4 GWP = 27.9, N_2O = 273). Select the version required by your national inventory reporting guidelines.

Random seed: Fixing the seed ensures results are exactly reproducible. Change the seed to verify convergence: if the 95% CI changes substantially between runs, increase the number of iterations.

Analysis mode: Single-year (most common) or Trend (multi-year time series).

Emission sources: Tick all sources applicable to your inventory. Unticked sources contribute zero emissions to the total.

7.2 Trend mode — year-to-year coefficient correlation

If you picked **Trend** analysis mode on Tab 1, an extra control appears on Tab 5: **Year-to-year correlation**. It governs how the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, ...) move from one year to the next *within a single Monte Carlo iteration.* Activity data (N, BW, Milk, ...) is always re-sampled fresh each year — this setting only affects the per-head coefficients.

Why does this matter? If your country uses the same emission factors for the whole inventory series (the typical case — either IPCC defaults or one national estimation), then a high-Ym draw in your base year should also be a high-Ym draw in the current year. The coefficient uncertainty then cancels when you compare the two years, so the trend uncertainty reflects only the activity-data changes. This is what IPCC V1 Ch3 §3.2.3 calls a "Type A'' sensitivity. If you instead treated EFs as independent across years, you would artificially inflate the trend uncertainty.

The three options:

- **Fully correlated coefficients (IPCC 2019 default)** — the same coefficient draw is reused for every year of the inventory within one MC iteration. *Pick this if your EFs are IPCC defaults or come from a single national estimation programme reused across the whole series.* This is the IPCC-recommended default and the conservative choice for trend uncertainty.
- **Partial (AR(1), $\rho = 0.7$)** — coefficient draws drift slowly between years (last year's value gets 70% weight, a fresh draw 30%). *Pick this if your EFs are re-estimated periodically — e.g.*

every 3–5 years — and neighbouring years share most of the same observational basis.

- **Independent (no year-to-year correlation)** — coefficient draws are sampled fresh each year. Maximises the EF contribution to trend uncertainty. *Pick this only if you genuinely re-measured every emission factor every year with fully new field data*, which is rarely realistic for a national inventory.

A practical sanity check: re-run your trend with `Fully correlated'' and again with`Independent''. If the two trend uncertainties are similar, your result is robust to this choice. If they are very different, your trend uncertainty is dominated by the EF block and you need to be explicit about which assumption you're making in your inventory documentation.

Trend mode runs n_iter simulations **per year**, so total compute = n_iter × number of years. Use 5{,}000–10{,}000 iterations for trend runs in development and 25{,}000–30{,}000 for final reporting.

7.3 Running the simulation

Click **Run Monte Carlo Simulation**. A progress bar shows the simulation steps. For 10,000 iterations and a single production system, typical run time is 15–60 seconds.

When complete, results appear in the lower section of the same tab.

If something is missing or not working: All settings (iterations, GWP, seed, emission sources) remain active in the upper section of Tab 5 — simply adjust any setting and click **Run Monte Carlo Simulation** again. The previous results are replaced automatically. If you need to change your input data (parameters, distributions, or correlations), go back to the relevant tab (Tab 1–4), make the change, then return to Tab 5 and re-run.

7.4 Convergence diagnostics

Click "Run Diagnostic" (magnifying glass button) after a simulation to assess whether enough iterations were run for stable results:

- **MCSE%** (Monte Carlo Standard Error %): Pass $< 0.5\%$, Warn $0.5-1\%$, Fail $\geq 1\%$
- **Mean drift %**: How much the mean from the first half of iterations differs from the second half. Pass $< 2\%$
- **CI drift %**: How much the 95% CI bounds shift between the two halves. Pass $< 5\%$
- **Convergence trace**: Running mean and 95% CI plotted against iteration count — should stabilise before the end of the run

Note: this tool uses **independent Monte Carlo**, not MCMC. Each iteration is drawn independently — there are no chains, no warmup, and no burn-in.

8 Understanding results (Tab 5 — Results)

8.1 Headline metrics

After a successful run, five value boxes show:

- **Total CO_2eq (t)**: The mean of the simulated total emission estimate, in tonnes CO_2-equivalent
- **Enteric CH_4 (t)**: Mean enteric fermentation contribution *plus* the source-specific 95% margin of error inline (e.g. `2,608 t · ±12%`)
- **Manure CH_4 (t)**: Mean manure CH_4 *plus* its 95% MoE
- **Pasture N_2O (t)**: Mean combined direct and indirect N_2O from pasture deposition *plus* its 95% MoE
- **Total 95% MoE (%)**: The headline uncertainty — half-width of the 95% confidence interval as a percentage of the mean. This is the IPCC reporting convention (Vol.1 Ch.3 Table 3.3) and the single uncertainty metric used across every user-facing display in the app (results tables, comparison chart, IPCC Annex 7, Word report). Coefficient of variation (CV%) is computed internally and exposed in the methodology document for context, but is no longer surfaced in any of the live tables to keep the user interface consistent.

Headline by cattle type. Directly below the value boxes, a "Headline by cattle type" card appears whenever your inventory contains more than one `cattle_type` on the Parameters sheet (typically `dairy` and `other`). It shows one row per cattle type with mean ± 95% MoE for each of the four headline sources plus the total CO_2eq. Single-cattle-type runs do not see this card.

8.2 Uncertainty decomposition (AD vs EF)

If "Run uncertainty decomposition (AD/EF/Combined)" was checked, the results include a bar chart with the 95% MoE on the y-axis (IPCC reporting convention) under three configurations:

- **AD uncertainty**: Driven by activity data only (N, BW, Milk, WG) with all emission factors fixed at their means
- **EF uncertainty**: Driven by emission factors and coefficients only with activity data fixed at means
- **Combined uncertainty**: Both sources varying together

If AD uncertainty dominates, improving livestock population surveys would most effectively reduce total uncertainty. If EF uncertainty dominates, investing in country-specific EF measurement studies would be more impactful.

8.3 By-system and by-category breakdown

- **By-system breakdown**: Mean, 95% CI, MoE%, and CV% for each production system and emission source. The aggregation-level selector at the top of the section lets you switch between `cattle_type` (default — dairy / other), `aggregation_level` (production system) and `sub_category`.

- **By reporting category (IPCC Table 3.3 layout)**: Grouped by IPCC source category (3.A.1, 3.A.2, 3.C.4, 3.C.5). The leftmost column is always `Cattle type`, regardless of the aggregation-level selector. Each row carries mean values for the raw gas (Mean (t CH_4) *or* Mean (t N_2O) — only one is populated per source) *plus* Mean (t CO_2eq) for cross-source comparison, plus MoE%, CV%, and the 95% CI bounds.

9 Sensitivity analysis (Tab 6)

9.1 Reading the tornado chart

The tornado chart ranks parameters by their contribution to total uncertainty. The longest bars indicate the parameters that, if better constrained, would most reduce the 95% CI.

Each bar is labelled as `<parameter> (<sub_category>)` — for example `Ym (DINT_cow)` vs `Ym (DINT_heif)`. The sub-category in parentheses identifies which animal group the influential parameter belongs to, so you can target research at the specific sub-category driving uncertainty rather than at the parameter in general. The same labels appear in the rank-correlation table, the Excel SRC / PRCC sheets, and the Word report's sensitivity section. Per-MMS sampled parameters carry their MMS in the parameter name and the sub-category in parentheses, e.g. `MCF_solid_storage (DINT_cow)` or `fraction_pasture (DINT_heif)`.

Per-source sensitivity. The "Output" dropdown above the tornado lets you re-rank parameters for a single emission source (Enteric CH_4, Manure CH_4, etc.). The per-source view aggregates across ALL sub-categories — earlier versions of the tool silently restricted the per-source view to the first sub-category only, dropping every other sub-category's contribution; that has been fixed.

Bars are coloured by reducibility:

- **Green**: Uncertainty can be reduced with better local data (surveys or measurement studies)
- **Grey**: Uncertainty is largely irreducible with national-level data (e.g. fundamental biochemical constants)

9.2 SRC vs PRCC

Two sensitivity methods are computed:

- **SRC** (Standardised Regression Coefficient): Linear sensitivity. Reliable when relationships are approximately linear.
- **PRCC** (Partial Rank Correlation Coefficient): Rank-based, captures non-linear relationships. More robust but less interpretable.

Both are shown; they typically agree on the top-ranked parameters. If they disagree significantly, the emission pathway may be strongly non-linear.

10 Generating the IPCC report (Tab 7 — IPCC Report)

10.1 IPCC Table 3.3

The IPCC Report tab shows your results formatted as IPCC 2006 Vol. 1 Ch. 3 Table 3.3. The three uncertainty columns in the downloadable Excel sheet are now labelled explicitly so there is no ambiguity:

- **AD uncertainty (95% MoE %)**: % uncertainty from activity data alone
- **EF uncertainty (95% MoE %)**: % uncertainty from emission factors alone
- **Combined uncertainty (95% MoE %)**: Overall % uncertainty

Definition of % uncertainty: Following IPCC Table 3.3 conventions, all values are expressed as the half-width of the 95% confidence interval divided by the mean, multiplied by 100 — i.e. `MoE% = ((CI_upper - CI_lower) / 2) / mean * 100`. CV (`SD/mean`) is a different metric and is **not** what IPCC Annex 7 expects in this column.

Example: a Combined uncertainty of 35% means the 95% CI is approximately ±35% of the mean emission estimate.

10.2 Downloads

Three download formats are available:

- **Excel report (.xlsx)**: Full workbook with run settings, IPCC summary table, uncertainty metrics, and sensitivity results (SRC and PRCC)
- **CSV report**: The IPCC Table 3.3 formatted table only, for easy import into inventory reporting software
- **Word report**: Narrative summary suitable for inclusion in the uncertainty section of your national inventory report

11 IPCC reporting conventions

11.1 Where uncertainty results go in the national inventory report

IPCC 2006 Vol. 1 Ch. 3 §3.3 describes the standard approach for Approach 2 (Monte Carlo):

1. Report the 95% confidence interval for each major source category and for the national total
2. Present the results in Table 3.3 format (included in the IPCC Report tab downloads)
3. Describe the Monte Carlo approach used (number of iterations, distributions chosen, correlation assumptions)
4. Document any parameters where country-specific values were used instead of IPCC defaults, including the justification

The methodology PDF (available on the Resources tab) provides a complete technical description of the equations, sampling approach, and convergence assessment that can be referenced in your national inventory report.

11.2 What IPCC expert reviewers will check

Based on experience with IPCC expert reviews, reviewers typically focus on:

- Whether the uncertainty metric reported is the correct one (% uncertainty = MoE%, not CV%)
- Whether the IPCC Table 3.3 format and column conventions are followed
- Whether Approach 2 (Monte Carlo) is described sufficiently to be reproducible
- Whether the number of iterations is sufficient for stable results
- The basis for any country-specific emission factor values that deviate substantially from IPCC defaults

12 Worked example — Country X (hypothetical dairy)

To illustrate the workflow, the following example uses the Country X built-in dataset.

Step 1 — Load data: Select "Country X (hypothetical dairy)" from the dropdown on Tab 1. The data loads automatically (12 parameters, dairy smallholder system).

Step 2 — QA/QC: Go to Tab 2. All parameters should pass; any Warn flags would compare against the IPCC 2019 Refinement defaults (Vol.4 Ch.10 / Ch.11), and a deliberate country-specific value can be documented in the inventory report rather than corrected.

Step 3 — Uncertainty: Go to Tab 3. Note that activity data (N, BW, Milk, WG, Fat) use Normal distributions and coefficients (Ym, DE, Bo, etc.) use PERT or lognormal distributions.

Step 4 — Correlations: Go to Tab 4. Select "From template (auto, time-series)" — the example dataset includes a 5-year time series from which a Spearman rank correlation matrix is computed automatically (first-difference detrending is applied internally to strip shared long-run growth). The matrix is sampled using Iman–Conover restricted pairing. Leave the right-hand card at "No EF correlations" for the default run.

Step 5 — Simulate: Go to Tab 5. Set iterations to 10,000, GWP to AR6, seed to 42. Click "Run Monte Carlo Simulation".

Step 6 — Results: After about 30 seconds, results appear. Note the 95% CI and MoE%. Click "Run Diagnostic" to confirm convergence.

Step 7 — Sensitivity: Go to Tab 6. The tornado chart shows which parameters drive most uncertainty. In this example, Ym and BW for cows are typically the dominant contributors.

Step 8 — Report: Go to Tab 7. Download the Excel report to see the full IPCC Table 3.3 output.

Funded by the Global Methane Hub.